Parental status and potential fertility stereotypes (#297542)

Author(s)
This pre-registration is currently anonymous to enable blind peer-review.
It has one author.

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1) Have any data been collected for this study already?
No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?
H1–H2 (Fertility Potential Penalty): Among both childless women and mothers, applicants with short-term potential fertility (plans to have children in 2 years) will receive worse evaluations than those with low (no plans) or long-term (5 years) potential fertility, on perceived cost, risk, and growth potential (H1, childless) and additionally competence (H2, mothers).
H3 (Motherhood Penalty): Mothers will receive worse evaluations than childless women on perceived cost, risk, growth potential, and competence, independent of fertility potential. Mothers with short-term potential fertility are expected to be the most penalized group across all conditions and measures.
H4–H9: All penalties described in H1–H3 are expected to be significantly smaller or absent for male profiles; men are not expected to face equivalent penalties for short-term potential fertility or fatherhood.
H10–H15 (Occupational Level Moderation): Occupational level (professional vs. technical) is expected to significantly moderate the effects in H1–H3, with direction varying by outcome: professional women are expected to face larger penalties on longer-term measures (growth potential, likelihood to quit, overall cost), while technical women are expected to face larger penalties on shorter-term measures (likelihood to be late/absent, competence, reliability). These hypotheses are confirmatory regarding the existence of moderation, without a uniform directional prediction across outcomes.
H16 (Single Motherhood Penalty): The motherhood penalty (H3) is expected to be larger for single than married mothers, with this marital-status gap specific to women (no equivalent gap expected among fathers).
All analyses not explicitly specified in this pre-registration are exploratory and will be clearly distinguished as such in all reporting.

3) Describe the key dependent variable(s) specifying how they will be measured.
Hiring recommendation (1 item): probability of recommending the applicant for hiring.
Third-order evaluations (8 items): how most people in the firm would evaluate the applicant on a set of measures: reliable, competent at work, committed to the firm long-term, professionally promising, ambitious about career growth, risky in terms of labor continuity, costly in terms of salary, costly in terms of operational or management burden, and a good investment (will generate more benefit than cost for the firm).
Predicted behavioral outcomes (6 items): probability that the applicant would: use firm benefits, request accommodations or leave, arrive late or be absent from work, resign or indefinitely leave the workforce, take on new challenges within the firm, and be promoted.
Items are theoretically grouped into four constructs: perceived cost, perceived risk, growth potential, and current competence. If internal consistency within each construct is adequate (Cronbach's alpha >= .70), items will be aggregated into composite indices computed as unweighted means. If the threshold is not met, items within that construct will be analyzed individually. This decision rule is pre-specified. All individual item analyses will be reported regardless of whether composite indices are used.

4) How many and which conditions will participants be assigned to?
Occupational level (professional vs. technical) is a between-subjects factor: each participant is assigned to evaluate profiles of either professional or technical applicants only, with mutually exclusive samples of ~800 participants per condition and no overlap between groups. Within their assigned condition, each participant evaluates exactly 2 profiles of the same occupational type, drawn from one pre-generated block.
Within each occupational condition, six attributes are manipulated at the profile level using a D-efficient design (200 unique profiles, 100 blocks of 2 profiles each): gender (2: woman/man), marital status (2: single/married), parental status (3: no children/1 child/2 children), family plans (3: plans in 2 years/5 years/no plans), experience (3: 6/7/8 years), and technical performance on a test (3: grades of 5.8/6.0/6.3). All attribute combinations are randomly distributed across participants via sequential block assignment. Each participant evaluates both profiles sequentially and independently on 7-point scales.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.
We will conduct regression models accounting for nested data structure, estimated separately for each outcome, each construct, and each type of occupation. Our analyses will account for the nesting of 2 profile-level observations within each respondent, as it is the standard method for factorial survey experiments (Hainmueller et al., 2014; Pustejovsky & Tipton, 2018).
Five confirmatory model sets, all controlling for presentation order and template version:
SET 2a: SET 1 + Gender × Parental Status (tests H3, H4–H9).
SET 2b: SET 1 + Gender × Family Plans (tests H1, H2, H4–H9).
SET 3: SET 1 + Gender × Marital Status (tests H4–H9).
SET 4a: SET 1 + Gender × Marital Status × Parental Status (tests H16).
SET 4b: SET 1 + Gender × Parental Status × Family Plans (tests H4–H9).
SET 4a and SET 4b probe two distinct moderators of the motherhood penalty (H3): marital status (H16) and future fertility plans (H1, H2), respectively.
The following confirmatory model sets are estimated on the pooled sample across both occupational conditions:
SET 5: SET 1 + Occupational Level (tests H10).

SET 5a: SET 1 + Occupational Level × Gender (tests H11).
SET 5b: SET 1 + Occupational Level × Parental Status (tests H12).
SET 5c: SET 1 + Occupational Level × Family Plans (tests H13).
SET 5d: SET 1 + Occupational Level × Gender × Parental Status (tests H14).
SET 5e: SET 1 + Occupational Level × Gender × Family Plans (tests H15).
Predicted marginal means estimated for theoretically specified group comparisons. To anchor the interpretation of the fertility- and motherhood-related penalties (H1–H3, H16), we will additionally estimate and report the predicted marginal means contrast between two reference profiles from SET 1: (a) male, single, no children, no family plans, and (b) female, single, no children, no family plans - i.e., identical profiles differing only in gender. This contrast provides a baseline estimate of any gender gap in evaluations that is independent of fertility-related attributes, against which the magnitude of H1–H3 and H16 effects can be compared. All other analyses are exploratory.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.
Data will be collected through Netquest, a commercial online panel provider through which the survey will be administered to participants in Chile. Netquest applies its own internal attention checks prior to survey completion; participants who fail these checks are excluded by the panel provider before appearing in the dataset. Our survey includes an additional task-specific attention check item asking participants to identify which piece of information was NOT included in the profiles (correct answer: RUT). Rather than excluding participants who fail this item, we will conduct all analyses both on the full sample and restricted sample, and report a sensitivity analysis comparing results with and without participants who failed the attention check, to assess whether their inclusion affects conclusions while preserving statistical power.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.
Approximately 800 participants per occupational condition (~1,600 total), yielding ~1,600 profile-level observations per occupational condition nested within respondents. Sample size is constrained by maximum panel availability for the target population (supervisory/managerial workers in Chile) through Netquest. The D-efficient design achieved D-efficiency = 0.922, and G-efficiency = .925, confirming adequate estimation efficiency for all specified effects. With 800 participants each evaluating 2 profiles, each of the 100 unique blocks is assigned to 8 participants.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)
Secondary analyses (clearly distinguished from confirmatory in all reporting): moderation by evaluator gender, parental status, and hiring experience; moderation by workplace norm attitudes. Any analysis not explicitly specified in this pre-registration, including analyses of individual outcome items outside the four theoretical constructs, interaction terms not specified in SET 1 through SET 5e, and any comparison not listed in the expected values section, will be treated as exploratory.

A pretest was conducted with 200 participants from the same target population prior to this registration to assess survey functionality, data quality, and potential design-related artifacts, including profile order and template effects. Based on the pretest, one structural design change was implemented: occupational level was moved from a within-subjects attribute to a between-subjects factor due to evidence that its presentation interacted with profile order. The pretest was used solely for instrument and design refinement and was not used to evaluate the direction, magnitude, or statistical significance of any confirmatory hypotheses. No pretest observations will be included in the confirmatory analyses, which will be conducted exclusively on data collected after this preregistration.